

Computer Vision
Assignment IV

Detection and tracking

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Motivation and Context

Object detection and tracking in aerial images (UAVs/Drones) are critical tasks in modern surveillance, traffic management, and smart city operations. Unlike standard datasets, aerial views present unique challenges: extreme variations in altitude, small object scales, high density of vehicles, and significant camera motion. Traditional tracking methods relied on simple geometric associations. However, modern approaches integrate Deep Learning detectors with sophisticated data association strategies. This project aims to explore the state-of-the-art (SOTA) in Multi-Object Tracking (MOT) applied to vehicular traffic by decoupling the detection and association phases.

Dataset Architecture and Protocols

The UAVDT-Benchmark (Test Subset) will be used. This specific subset (1.2 GB) is mandatory for all groups, to compare the results and eliminating the need for long training phases. This dataset follows a strict MOT hierarchy that students must respect for a valid evaluation.

Folder Structure

Assuming the dataset is exported in standard MOT format (e.g., via Dataset Ninja), the following structure will be found:

- `sequences/`: Contains subfolders for each video sequence (e.g., `M0203`, `M0205`).
- `img1/`: Located inside each sequence folder. Contains sequentially numbered JPEG images (e.g., `000001.jpg`, `000002.jpg`). These are the input data for the pipeline.
- `gt/`: Located inside each sequence folder. Contains the Ground Truth file (`gt.txt`). This manually constructed label is used exclusively for metrics calculation.
- `det/` (Ignore if present): Contains pre-calculated detections using legacy models. These must not be used. The core of the project is to generate new detections using modern models.

MOT Challenge Annotation Format

Both the `gt.txt` and the students' prediction results must follow the MOT standard:

`<frame>, <id>, <bb_left>, <bb_top>, <bb_width>, <bb_height>, <conf>, <class>, <vis>`

1. `frame`: Frame index (starts at 1).
2. `id`: Unique trajectory identifier (must be persistent).
3. `bb_left, bb_top`: (x, y) coordinates of the top-left corner.
4. `bb_width, bb_height`: Bounding box dimensions in pixels.
5. `conf`: Detector confidence (usually 1 in the GT).
6. `class`: Object category. Restriction: Filter results exclusively for vehicular classes (Car, Truck, Bus). Ensure your model's class IDs map correctly to these vehicles.
7. `vis`: Visibility ratio (can be ignored or set to 1 in predictions).

Methodology

Students must implement a modular pipeline. The use of "all-in-one" functions (such as `model.track()`) is not allowed without a detailed justification of the component integration.

Model Comparison

Four different configurations must be compared:

- Method 1 (Baseline): YOLOv11 + SORT. A purely geometric approach.

- Method 2 (Efficiency): YOLO26 + ByteTrack. Analysis of NMS removal on tracking latency.
- Method 3 (Accuracy): RT-DETR + BoT-SORT. Use of global attention and Global Motion Compensation (GMC).
- Method 4 (Resolution): Any of the above integrating SAHI (Slicing Aided Hyper Inference) to detect extremely small vehicles, and you can also present beside this model, other heuristic that improve your results.

Note that because of time and hardware constraints, students are not required to train models from scratch. The task requires a two-step inference process: first, detection (using pre-trained weights), and then bounding box matching. Since the images are from drones, standard COCO-trained algorithms may fail on small vehicles. Therefore, you must search online (e.g., HuggingFace, Ultralytics) for models pre-trained specifically on drone datasets (like VisDrone or UAVDT) to act as your robust detector. Furthermore, tuning the tracking algorithm's hyperparameters (like matching thresholds) is crucial here. The challenge of small object detection without re-training will be fully demonstrated in the fourth method using SAHI.

Evaluation Protocol

Quantitative evaluation will be performed using the `motmetrics` library. The following averaged indicators must be reported across all test sequences:

- MOTA (Multiple Object Tracking Accuracy): Accumulated error of false positives, false negatives, and ID switches.
- IDF1 (ID F1-score): The model's ability to maintain identity over the long term.
- ID Switches (IDS): Total number of times a vehicle changes its identifier.
- FPS (Frames Per Second): Temporal performance of the entire system (Inference + Association).

Deliverables

An oral classroom presentation supported by slides, including:

- Technical explanation of the YOLO26 vs. RT-DETR architecture.
- Justification for using appearance (embeddings) versus purely geometry in tracking.
- Analysis of the impact of SAHI on low-resolution vehicle detection.
- Visual demos showing success cases and failures (e.g., occlusions caused by trees or bridges).

Rubric

For this practice, only the presentation is required; a written document is not necessary. The rubric is as follows:

- (5pts) Presentation: Slide content and formality.
- (7pts) Knowledge: Semantic content of the slides.
- (3pts) Q&A: Questions answered during the presentation.
- (5pts) Innovation: Heuristics or parameter tuning used to improve results.